AYUSHI

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SUMMARY

AI/ML Engineer with a focus on Machine Learning and Full-Stack Integration. Specialized in developing high-performance models using XGBoost and CatBoost and deploying LLM applications via RAG pipelines. Demonstrated expertise in achieving 90%+ prediction accuracy and optimizing backend systems for a 40% reduction in API latency. Experienced in multilingual NLP, RESTful API design, and scalable MERN/Flask architectures.

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, C, SQL
- **Generative AI & NLP:** LLaMA 3.1, Prompt Engineering, RAG Pipelines, Groq API, gTTS, Multilingual NLP
- **Machine Learning:** Scikit-learn, XGBoost, CatBoost, Random Forest, KNN, Ensemble Learning.
- **Web Development:** Flask, Node.js, Express.js, HTML5, CSS3, RESTful APIs
- **Data & Infrastructure:** Pandas, NumPy, MongoDB, MySQL, Git/GitHub, Gradio, HuggingFace Spaces

PROJECTS

Nyaya Mitra – AI Legal Document Assistant **[GitHub]** | *Mar 2026*

- Engineered a multilingual legal processing tool using Llama 3.1 to parse and simplify FIRs and court notices for rural populations.
- Architected a structured data extraction pipeline that normalizes dense legal jargon into JSON-formatted actionable insights.
- Deployed a speech synthesis (TTS) framework supporting 4+ regional languages to improve accessibility for semi-literate users.
- **Tech Stack:** Python, Groq API, Gradio, PyPDF2, gTTS, HuggingFace.

Crop Recommendation System – Full-Stack Application **[GitHub]** | *Apr 2025*

- Developed a 3-tier system utilizing Flask for model inference and Node.js for backend API orchestration.
- Validated model performance with 90%+ accuracy using soil and meteorological features (NPK, pH, Rainfall).
- Optimized a MongoDB data pipeline, achieving a 40% reduction in query response times.
- Integrated the OpenWeatherMap API to provide real-time agricultural analytics and improved UI responsiveness.
- **Tech Stack:** Python, Scikit-learn, Flask, Node.js, Express, MongoDB, OpenWeatherMap API.

AquaPredict – Rainfall Forecasting *Jan 2025*

- Trained an ensemble model (XGBoost, CatBoost, Random Forest, KNN) achieving 90% prediction accuracy.
- Preprocessed a dataset of 50,000+ meteorological records, reducing model training time by 40% through feature engineering.
- Utilized Python, Scikit-learn, and Pandas for high-dimensional data cleaning and model evaluation[cite: 77].
- **Tech Stack:** Python, Scikit-learn, XGBoost, CatBoost, Pandas, NumPy.

EDUCATION

Vellore Institute of Technology, Bhopal *Expected 2027*
B.Tech in Computer Science (Specialization in AI & ML) **CGPA: 8.7/10**

Kendriya Vidyalaya, Katihar *2022*
XII (CBSE) **84.6%**

Scottish Public School, Katihar *2020*
X (CBSE) **92.4%**

CERTIFICATIONS

- **Introduction to Internet of Things** – NPTEL (May 2026)
- **Applied Machine Learning in Python** – University of Michigan, Coursera (Dec 2025)
- **Cloud Computing** – NPTEL (May 2025)